CS 4483
Deliverable 2: Mid-Project Report
Escape the Island

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1 Level Design & Player Guidance

The Unity prototype implements a simplified open-island layout designed to prioritize clarity, accessibility, and iterative development. The environment consists of a navigable plane populated with resource nodes and a fixed raft construction location, allowing players to explore freely while maintaining a clear objective.

Player guidance is achieved through environmental placement and UI feedback. Resource objects are distributed across the island to encourage exploration, while the raft acts as a visible anchor point representing the player's long-term goal. The UI continuously displays resource counts and progression, reinforcing player direction without requiring explicit instructions.

The day–night cycle further structures player behavior. During the day, players are encouraged to gather resources, while the night phase introduces risk through an active enemy that pursues the player. This creates a natural guidance system based on shifting gameplay conditions rather than forced tutorials.

2 Narrative Design & World-Building

The narrative design of the prototype focuses on minimal but effective environmental storytelling. The core premise of being stranded on an isolated island is conveyed through gameplay rather than explicit narrative systems.

The survival loop itself communicates the narrative: players must gather resources, avoid danger, and work toward escape. The day–night cycle reinforces thematic tension, with daytime representing safety and productivity, and nighttime introducing uncertainty and threat.

Although dialogue systems and detailed lore are not yet implemented, the prototype establishes a clear thematic foundation aligned with the intended survival and light horror experience. Future iterations will expand on this through environmental storytelling elements and additional narrative mechanics.

3 Systems Design & Game Balance

The prototype integrates several core gameplay systems, including player movement, resource collection, a resource economy, a construction system, and a time-based enemy threat.

The resource system is centered around wood collection. Players gather resources by interacting with environmental objects, which are then stored and used to contribute toward raft construction. This creates a direct and intuitive gameplay loop.

Game balance is maintained through simple constraints. Resource collection requires movement and time investment, while progression requires multiple resources per upgrade. This prevents rapid advancement and encourages deliberate player decision-making.

The day–night system introduces dynamic difficulty. During the night phase, an enemy entity actively pursues the player, increasing pressure and reinforcing a risk–reward structure. Players must balance efficient resource gathering with survival considerations.

4 Progression Systems & Rewards

Progression is structured around the gradual construction of the raft, which serves as the primary long-term objective. Players contribute collected resources to incrementally increase raft completion.

The progression system operates across three levels:

- **Short-term:** Collecting individual resources
- **Mid-term:** Accumulating resources to contribute to raft progress
- **Long-term:** Completing the raft and achieving escape

This structure supports both intrinsic and extrinsic motivation. Players are intrinsically motivated to optimize their strategy and survive, while extrinsic motivation is reinforced through visible progression indicators such as percentage completion.

5 Testing & Iteration

Development involved continuous testing and rapid iteration during implementation. Initial versions of the prototype revealed several usability challenges, particularly in player interaction and system clarity.

One major issue was the lack of feedback when collecting resources or contributing to the raft. This was resolved by implementing UI elements that display real-time updates, improving clarity and player understanding.

Another issue involved the reliability of collision and interaction systems. Early inconsistencies were addressed by refining collider configurations and ensuring proper use of Unity's physics components.

The day–night system also required iteration to clearly communicate its impact. A visible indicator was added to display the current state, ensuring players can anticipate changes in difficulty.

These iterations improved overall usability, system reliability, and gameplay clarity.

6 Technical Implementation

The prototype was developed in Unity using C# scripts to implement core gameplay systems. Key components include player movement, resource collection, raft construction, enemy behavior, and the day–night cycle.

A centralized **GameManager** handles shared game state, including resource counts and progression tracking. This approach simplifies communication between systems and ensures consistency.

The day–night system is implemented using a timer-based toggle, switching between states at fixed intervals. Enemy behavior is conditionally activated during the night phase, reinforcing the gameplay loop.

UI elements were implemented using TextMeshPro to display resource counts, progression, and current game state. This significantly enhances player feedback and usability.

The Unity project is organized to include only essential directories (Assets, Packages, ProjectSettings), and a functional Windows build has been generated to allow the game to run independently of the Unity editor.

7 Core Systems and Gameplay Loops

7.1 System Overview

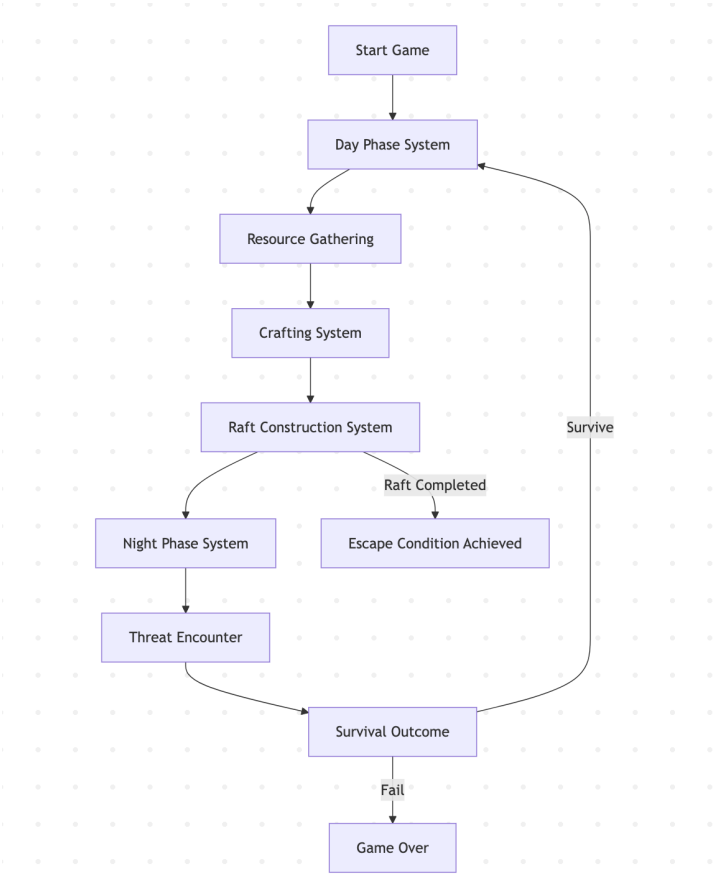

Figure 1: High-level system architecture of Escape the Island.

7.2 Raft Construction Progression

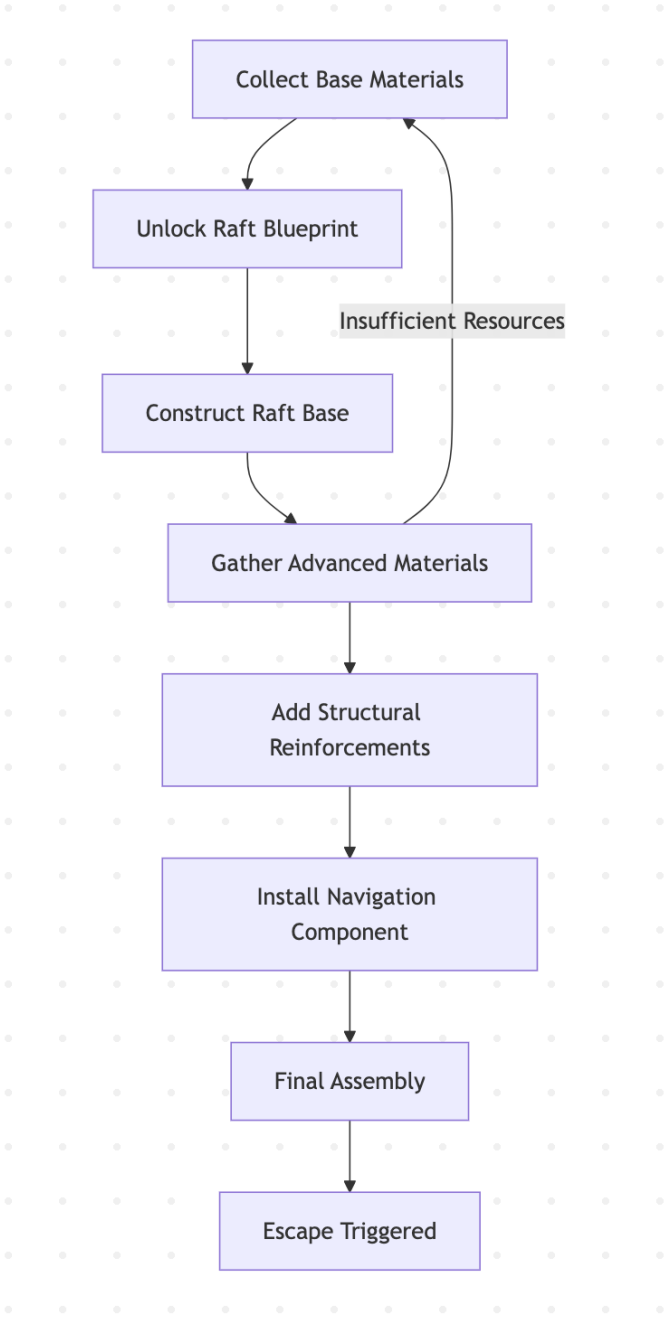

Figure 2: Structured meta-progression loop for raft construction.

7.3 Night Threat Subsystem

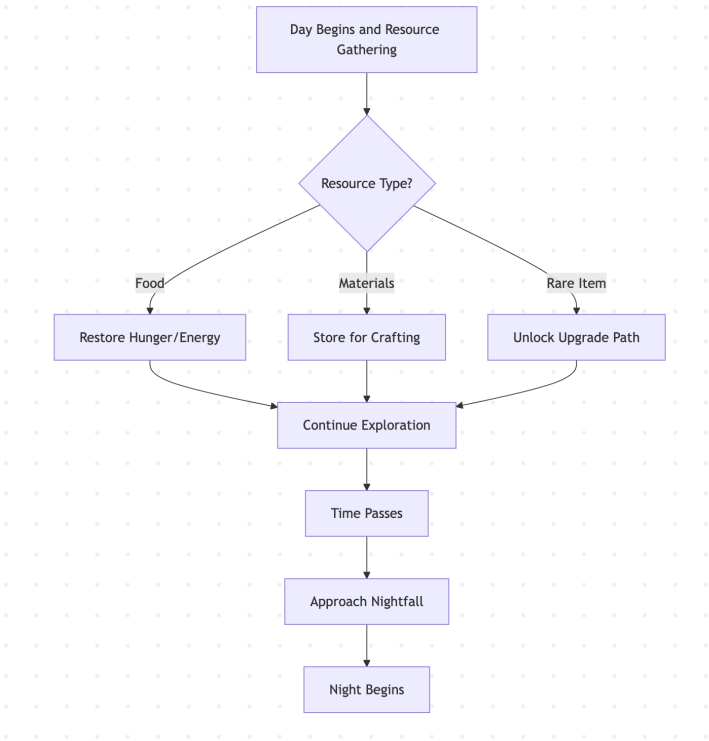

Figure 3: Night-time enemy behavior loop.

7.4 Day Survival Loop

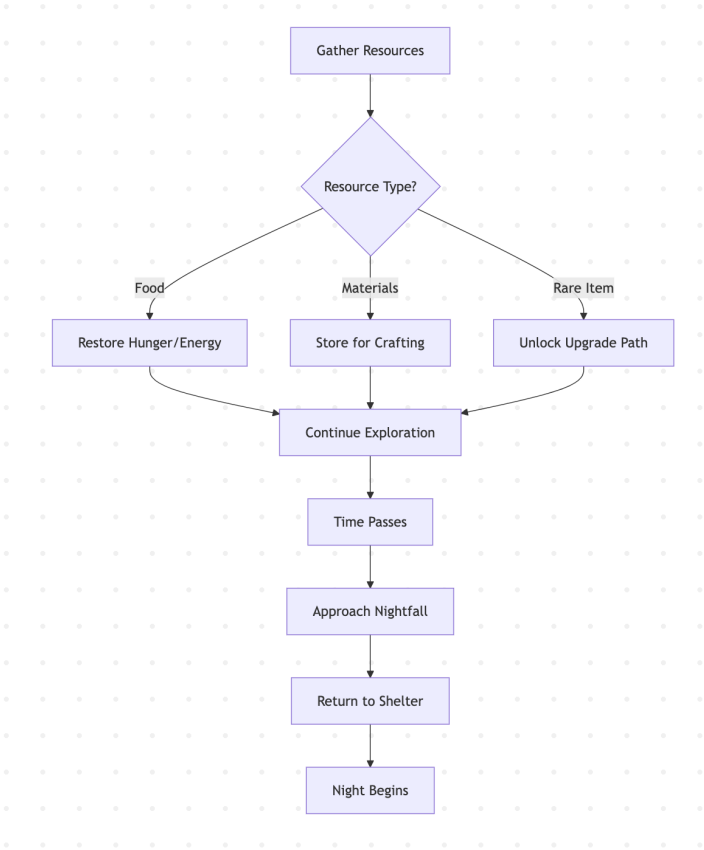

Figure 4: Daytime resource gathering loop.

7.5 Progression Loop

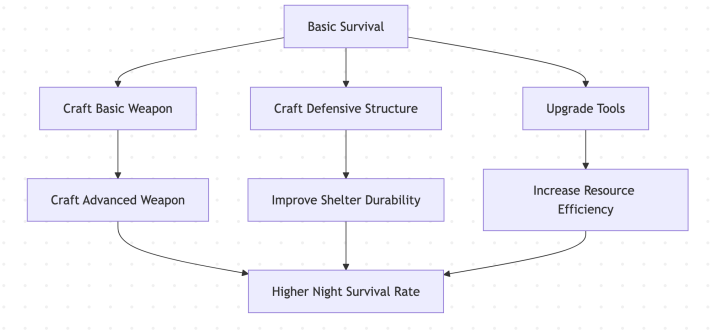

Figure 5: Resource-to-progression feedback loop.